

PROFESSIONAL SUMMARY

Full-stack software developer with practical, hands-on experience across the full technology set this role requires — Next.js (App Router), TypeScript, Tailwind CSS, React Native / Expo, Convex, Better Auth, Resend, WhatsApp messaging, Bunny.net CDN, Vercel + EAS — plus Node.js APIs, SQL and document-style data work, secure auth (JWT / RBAC), testing, and CI/CD. Also strong in Java, PHP, MySQL, Prisma, Azure, and AWS. Currently lead developer on FairBanks Medical Centre's FCHIP platform (community health intelligence, AI/ML, and related hospital systems). Previously shipped school, clinical, and embedded health software while leading biomedical programmes in accredited hospital settings. Ready to own web + mobile product work, secure APIs, and third-party integrations in a small, high-impact tech team.

TECHNICAL SKILLS

AgroMavericks-required stack (practical experience): Next.js (App Router), TypeScript, Tailwind CSS, React Native / Expo, Convex, Better Auth, Resend, WhatsApp messaging, Bunny.net CDN, Vercel, EAS

Full-stack delivery: React, Node.js / Express, HTML/CSS, Jest, Playwright, REST APIs, JWT, RBAC, CI/CD, automated testing

Backend, data & cloud: MySQL / SQL, Prisma, MongoDB-style document modelling, PHP (CodeIgniter, Laravel), Java, AWS, Microsoft Azure

AI & product engineering: AI agents, ML-assisted features, Cursor / OpenAI / Claude-assisted delivery; hospital management and HealthTech workflows

SOFTWARE EXPERIENCE

Lead Software Developer — FairBanks Medical Centre (FCHIP)

Current

- Lead development of FCHIP (FairBanks Community Health Intelligence Platform): AI/ML-assisted community health intelligence on top of live clinic and outreach operations.
- Design, build, and deploy hospital / clinic software including hospital management system workflows used in real care settings.
- Ship full-stack features across web and mobile using the modern product stack (Next.js, TypeScript, Tailwind, React Native / Expo, Convex, auth, messaging, CDN, and cloud deploy) plus cloud storage and data pipelines.
- Build and operate AI agents for production use cases (including a fully functioning AI agent deployed for a law firm in Texas, USA).

SELECTED SOFTWARE PROJECTS

Shulekeeper — school information system

Assistant Software Developer · 2021–2024

- Contributed to end-to-end web delivery: React / Next.js frontend with TypeScript, Laravel backend, MySQL data model, and AWS hosting.
- Translated business needs into requirements, process flows, and usable UI; integrated components and third-party programmes; supported testing and upgrades.
- Trained users and produced manuals — experience owning product beyond code alone.

Endoscopy reporting software

Lead Software Developer · St. Catherine Hospital & St. Francis Hospital · 2017–2019

- Led planning, algorithms, JavaFX UI, and MySQL relational design for medical image capture and reporting on endoscopy systems.
- Migrated existing user data, installed and configured software on multiple systems, and supported troubleshooting and upgrades.

Wekebere — fetal heart-rate monitoring

Electronics Developer & Programmer · 2017–2024

- Built Android (Java) mobile app linked to Arduino-based hardware for third-trimester fetal heart-rate monitoring; used Microsoft Azure in the stack.
- Designed system flow between hardware and app — useful background for mobile + device / field integrations.

HEALTHCARE OPERATIONS LEADERSHIP (SELECTED)

Biomedical Programs Manager — Gould Family Foundation (GFF)

Aug 2024 – Feb 2025

- Led biomedical programmes across multiple healthcare facilities; owned equipment lifecycle strategy, procurement, installation, and commissioning.
- Trained engineers and technicians; enforced international healthcare technology standards.

Biomedical Manager — International Hospital Kampala (IHK)

Jan 2020 – Jan 2024

- Managed biomedical operations for a major private hospital; led CT, oxygen plant, X-ray, and ICU technology projects.
- Built preventive maintenance systems; supported successful COHSASA accreditation and continuous quality compliance.

Biomedical Engineer — Norvik Hospital Ltd

Apr 2019 – Jan 2020

- Maintained and serviced diagnostic and monitoring equipment; supported installation, calibration, and preventive maintenance programmes.

EARLIER ROLES

Research Intern — Uganda Virus Research Institute

Dec 2018 – Apr 2019

- Biomedical research support and laboratory data analysis.

Teaching Assistant — Makerere University, College of Health Sciences

Mar 2016 – Aug 2017

- Supported biomedical engineering teaching and laboratory practicals.

EDUCATION

University of Washington

Certificate — Leadership and Management in Health · 2022 (A+)

Makerere University

BSc Biomedical Engineering (Second Upper Honours) · 2012–2017 · Key courses: ICT, OOP, Software Engineering, Database Systems

Green Bridge School of Open Technology

Advanced Java Programming · 2016 (A+)

REFEREES

Eng. Richard Ssejongo — Biomedical Engineer, St. Francis Hospital Nsambya · +256 753 818 754 / +256 777 132 489

Dr. Annet Khingi — Senior Radiologist & Administrator, Mengo Hospital · +256 772 592 771 / +256 701 592 771

Dr. Mamello Muhanuuzi — Former Medical Director, International Hospital Kampala · +256 702 811 008

Portfolio / GitHub links: available on request (add preferred URLs before submit).